

# Beyond Meta-Nash: Defensive PSRO for Robust Learning under Non-Transitive Competition

Anonymous submission

## Abstract

Real markets do not play Nash. Operators undercut prices to capture share, probe demand at a loss, or absorb short-term inefficiency for long-term market position. Policy Space Response Oracles (PSRO), the standard tool for non-transitive multi-agent learning, best-responds to a meta-Nash mixture over the learned population. When the real opponent steps outside that mixture, the trained agent has no defense. Street-level competitive ride-hailing with company-operated autonomous vehicle fleets is a canonical instance: per-vehicle control over location, pricing, and service makes off-equilibrium behavior an everyday strategy rather than a corner case. We propose Defensive PSRO (DF-PSRO). Instead of best-responding to the meta-Nash mixture, DF-PSRO reweights the opponent population via soft KL-DRO, shifting probability mass toward opponents that hurt the population’s strongest player. Under a mild reference-alignment condition, DF-PSRO provably dominates standard PSRO on worst-case payoff against any KL-bounded opponent deviation. To evaluate the framework, we build a microscopic Manhattan ride-hailing simulator calibrated on NYC TLC data and train agents with a hybrid policy that combines feature-engineered spatial priors for stability with a U-Net residual controller for fine-grained local refinement. Across aggressive price undercutting, brand-biased demand, and fleet-size asymmetry, DF-PSRO matches standard PSRO when opponents behave, and significantly outperforms it when they don’t.

## Introduction

In traditional human-driven ride-hailing systems, platforms influence driver behavior indirectly through pricing, dispatching, and incentives. The emergence of company-operated CAV fleets fundamentally changes this paradigm: operators now directly control each vehicle’s location, pricing, and service decisions. This shift transforms competition from coarse zone-level supply–demand balancing into a street-level strategic interaction problem, where payoffs are determined by fine-grained spatial configurations, including proximity between vehicles and passengers, local competitor density, and micro-scale demand variations.

While this transition increases observational resolution, its more profound impact lies in altering the structure of the learning problem. At street level, strategies that perform well in one spatial configuration often fail in another, and no single policy dominates across all scenarios. Instead, compe-

tition induces a diverse set of qualitatively distinct behaviors—aggressive price undercutting, hotspot concentration, avoidance of competitor-dense regions, conservative relocation, and exploratory randomization. These behaviors interact in a non-transitive manner, where each strategy can be exploited by another. As a result, optimizing a single stationary policy is structurally insufficient: any fixed strategy admits a strategically harmful counter-response.

A natural solution is to learn a population of policies that collectively cover the strategic landscape. PSRO provide a principled framework for this setting by iteratively computing best responses to a meta-strategy over the current policy population, enabling approximation of complex non-transitive equilibria. However, standard PSRO implicitly assumes that opponents play according to the equilibrium mixture. This assumption is often violated in competitive ride-hailing markets, where operators frequently adopt off-equilibrium strategies, such as aggressive price undercutting to capture market share, overpricing to probe demand elasticity, or heuristic relocation patterns driven by incomplete information.

This mismatch between equilibrium-based learning and adversarial real-world competition leads to a critical failure mode: policies trained against the meta-Nash mixture can be systematically exploited by opponents whose behavior lies outside the equilibrium support. Importantly, such deviations are not rare anomalies but intrinsic features of competitive markets. Consequently, standard PSRO optimizes for average-case performance under equilibrium play, while neglecting robustness to strategically adverse opponents. A straightforward remedy is to adopt a maximin objective, optimizing performance against the worst-case opponent in the population. However, maximin training is often overly conservative: by focusing exclusively on the worst-case scenario, it sacrifices performance in typical settings and fails to exploit structure in the broader opponent distribution. This reveals a fundamental trade-off: PSRO is efficient but brittle, while maximin is robust but pessimistic.

In this paper, we address this gap by introducing Defensive PSRO (DF-PSRO), a robustness-enhanced variant of PSRO that interpolates between these two extremes. Instead of best-responding to the equilibrium mixture or a single worst-case opponent, DF-PSRO reweights the opponent population to emphasize strategically adverse policies, while

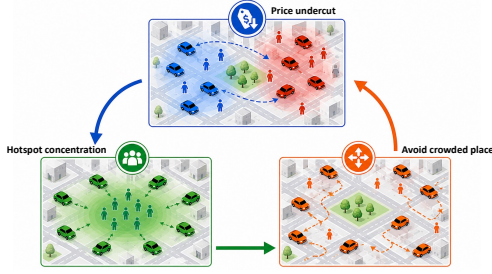


Figure 1: Local interactions over spatial configurations induce non-transitive competition, where no single policy is universally optimal. This naturally motivates population-based approaches such as PSRO, with street-level ride-hailing serving as a representative instance.

retaining diversity across the population. This reweighting admits a natural interpretation as a distributionally robust optimization (DRO) problem over the opponent strategy distribution, where a temperature parameter controls the trade-off between efficiency and robustness.

Our contributions are threefold:

- **Street-level competitive formulation.** We model competitive ride-hailing with CAV fleets as a street-level Markov game, where vehicle-level control induces non-transitive strategic interactions that cannot be captured by zone-level abstractions.
- **Defensive PSRO algorithm with theoretical guarantees.** We propose DF-PSRO, a population-based learning framework that performs soft KL-based reweighting over opponent strategies, and show that it provides a principled interpolation between standard PSRO and maximin optimization, with formal robustness guarantees.
- **Empirical validation under adversarial competition.** Using a microscopic ride-hailing simulator calibrated with real-world data, we demonstrate that DF-PSRO matches standard PSRO in normal settings while significantly improving performance under adversarial conditions, including aggressive price undercutting, demand shifts, and fleet asymmetry.

Together, these results suggest that robustness to off-equilibrium strategies is essential for learning in competitive multi-agent systems, and that population-based methods must explicitly account for adversarial deviations to remain effective in real-world markets

## Related Work

Learning-based approaches to ride-hailing and autonomous mobility-on-demand systems (Chen et al. 2019; Li et al. 2025; Toft et al. 2026) typically model competition at the zone or graph level, where spatial aggregation reduces the joint strategy space and simplifies learning, paralleling state abstraction in single-agent RL (Li, Walsh, and Littman 2006; Abel, Hershkowitz, and Littman 2016). While effective for large-scale coordination, such abstractions suppress fine-grained spatial interactions that become critical

when fleets are directly controlled. In CAV-based systems, local factors such as proximity to rival vehicles, block-level congestion, and demand imbalance directly determine outcomes, giving rise to non-transitive strategic behaviors that cannot be captured by coarse representations.

Population-based methods, particularly Policy Space Response Oracles (PSRO) (Lanctot et al. 2017), provide a natural framework for handling such non-transitive interactions by maintaining a policy population and iteratively computing best responses to a meta-strategy. Subsequent work improves the meta-solution concept, including correlated equilibrium in JPSRO (Marris et al. 2021) and  $\alpha$ -rank in  $\alpha$ -PSRO (Muller et al. 2020), as well as the training procedure through scalable or diverse best-response computation (McAleer et al. 2020, 2022; Liu et al. 2021). However, these approaches fundamentally optimize against a meta-strategy supported on equilibrium mixtures of the current population. While suitable for equilibrium analysis, this assumption is often violated in competitive markets, where agents adopt off-equilibrium strategies such as aggressive price undercutting or heuristic relocation. As a result, policies trained under the meta-Nash mixture may perform well in expectation but remain vulnerable to strategically adversarial opponents, even when such behaviors are already represented in the population.

Robust decision-making provides a natural direction for addressing this limitation. In single-agent settings, robust MDPs (Iyengar 2005; Nilim and El Ghaoui 2005) and their RL extensions (Smirnova, Dohmatob, and Mary 2019) optimize worst-case performance under model uncertainty, while risk-sensitive formulations such as CVaR (Chow et al. 2015; Tamar, Glassner, and Mannor 2015) emphasize low-return outcomes. Distributionally robust optimization (DRO) (Ben-Tal et al. 2013; Duchi and Namkoong 2021) generalizes these approaches by optimizing against worst-case distributions within an ambiguity set. However, these methods primarily address environmental uncertainty, rather than strategic uncertainty induced by adaptive opponents, and do not directly extend to population-based multi-agent learning. In particular, they do not specify how robustness should be incorporated into the meta-strategy over a policy population. This gap motivates our approach: incorporating distributional robustness into PSRO by reweighting the opponent distribution toward strategically adverse policies, enabling robustness to off-equilibrium strategies while preserving the benefits of population-based learning.

## DF-PSRO Algorithm

We model competitive ride-hailing with company-operated CAV fleets as a partially observable Markov game. Each company controls a fleet of vehicles operating over a shared road network and seeks to maximize cumulative revenue over a finite horizon. At each decision epoch, companies act based on partial observations of the system state, including the spatial distribution of vehicles, competitors, and traveler demand. Each action consists of operational decisions such as pricing, relocation, and service assignment for individual vehicles.

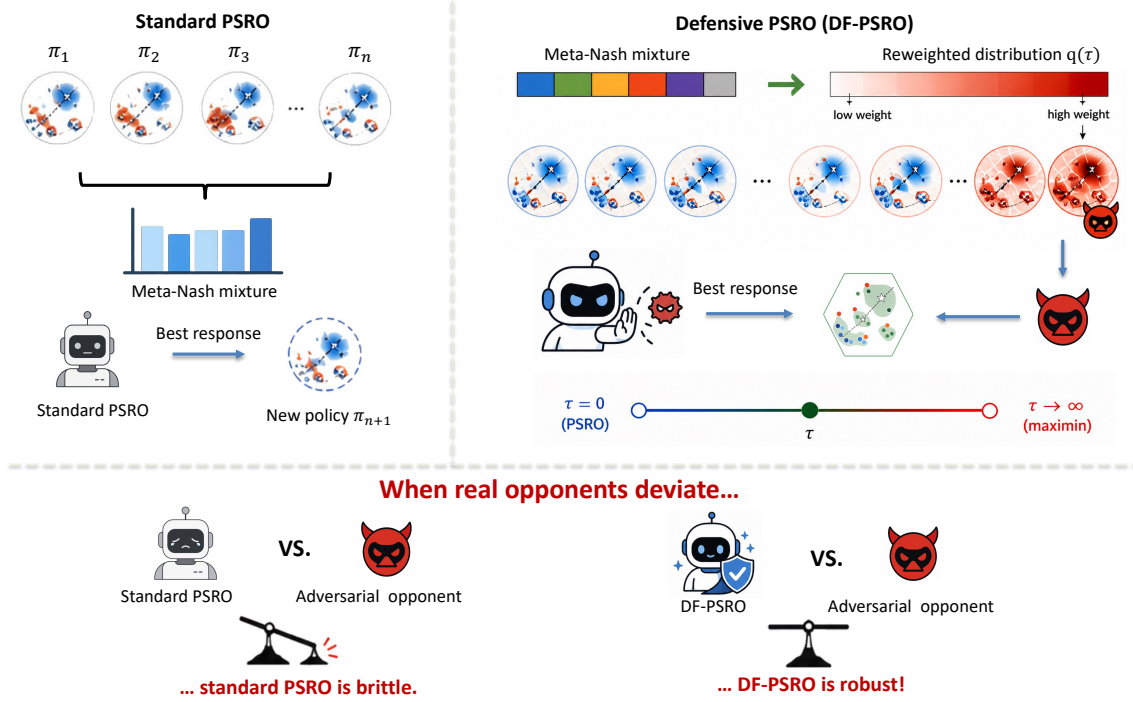


Figure 2: Overview of DF-PSRO.

The reward of each company is defined as the total revenue collected from completed trips. The performance of a policy therefore depends on the strategies adopted by competing companies, leading to a multi-agent learning problem over a distribution of opponent behaviors. Detailed definitions of observations, actions, transition dynamics, and reward construction are provided in Appendix .

Without loss of generality, we focus on the two-company case ( $K = 2$ ). At iteration  $t$ , each company  $i \in \{1, 2\}$  maintains a finite population of coordinator policies  $\Pi_i^t = \{\mu_i^1, \dots, \mu_i^{k_i}\} \subset \Pi_i$ . The induced *meta-game* is the finite normal-form game with payoff matrix  $M^t \in \mathbb{R}^{|\Pi_1^t| \times |\Pi_2^t|}$ , where  $M_{jk}^t = R_1(\mu_1^j, \mu_2^k)$ . A *meta-strategy*  $\sigma_i^t \in \Delta(\Pi_i^t)$  is a probability distribution over the current population; we write  $R_i(\mu_i, \sigma_{-i}^t)$  for the expected payoff of company  $i$  when playing  $\mu_i$  against opponents sampled from  $\sigma_{-i}^t$ .

**Maximin-PSRO.** A natural first step toward robustness is to replace the expected best response with a *maximin* objective that maximizes the worst-case payoff over the current opponent population:

$$\mu_i^{\text{new}} = \arg \max_{\mu_i \in \Pi_i} \min_{\mu_{-i} \in \Pi_{-i}^t} R_i(\mu_i, \mu_{-i}). \quad (1)$$

In training process, this oracle optimizes against an adversarially selected opponent from  $\Pi_{-i}^t$ : at each step, gradient signal concentrates on whichever  $\mu_{-i}$  currently yields the lowest payoff for  $\mu_i$ . Maximin-PSRO provides an explicit lower-bound guarantee on payoff regardless of which opponent in  $\Pi_{-i}^t$  is encountered. However, it optimizes an *absolute* worst-case payoff, which can be overly conservative: in

markets where all opponents are weak, the maximin solution may settle for a low but “safe” payoff and forgo achievable revenue improvements.

**DF-PSRO.** DF-PSRO replaces the meta-Nash mixture  $\sigma_{-i}^t$  with a reweighted opponent distribution  $q(\tau)$  that up-weights opponents capable of defeating strong policies in the current population. The overview of DF-PSRO is shown in Figure 2.

The reweighting follows a soft KL-DRO objective over the opponent population:

$$q_j(\tau) \propto \sigma_{-i,j}^t \cdot \exp(-\tau \cdot \tilde{R}_{bj}), \quad \tilde{R}_{bj} = \frac{R_{bj} - \bar{R}_b}{s_{R_b}}, \quad (2)$$

where  $b = \arg \max_i \sum_j \sigma_{-i,j}^t R_{ij}$  is the strongest incumbent in  $\Pi_{-i}^t$  under  $\sigma_{-i}^t$ ,  $R_{bj} = R_i(\mu_i^b, \mu_{-i}^j)$  is its payoff against opponent  $\mu_{-i}^j$ , and  $\bar{R}_b, s_{R_b}$  are the mean and standard deviation of  $\{R_{bj}\}_{j=1}^n$ . The temperature  $\tau \geq 0$  controls the robustness–efficiency trade-off, and the standardization makes  $\tau$  scale-invariant across iterations.<sup>1</sup>

The new best response is then trained against the reweighted distribution:

$$\mu_i^{\text{new}} = \arg \max_{\mu_i} \mathbb{E}_{j \sim q(\tau)} [R_i(\mu_i, \mu_{-i}^j)], \quad (3)$$

<sup>1</sup>Eq. (2) can be viewed as a tractable surrogate for full KL-DRO with  $\tilde{R}_{bj}$  replaced by the BR’s own payoff  $\tilde{R}(\mu_i, \mu_{-i}^j)$ , which would require re-evaluating  $\mu_i$  against the population at every gradient step. Anchoring to the strongest incumbent  $\mu_i^b$  preserves the PSRO per-iteration cost of standard PSRO.

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**Algorithm 1: Defensive PSRO (DF-PSRO)**


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**Require:** Initial policy  $\mu_0$ ; meta-solver  $\mathcal{M}$ ; temperature  $\tau \geq 0$ ; total iterations  $T$ ; best-response oracle  $\text{BR}(\cdot)$

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0: Initialize population  $\Pi \leftarrow \{\mu_0\}$ 
0: Initialize empirical payoff matrix  $M$  via self-play of  $\mu_0$ 
0: for  $t = 1, \dots, T$  do
0:   Compute meta-Nash mixture  $\sigma \leftarrow \mathcal{M}(M)$  over  $\Pi$ 
0:   Identify the strongest incumbent  $\mu_b$  in  $\Pi$  under  $\sigma$ 
0:   Extract and standardize  $\mu_b$ 's payoff vector against  $\Pi$ 
0:   Compute reweighted opponent distribution  $q(\tau)$  via
     soft KL-DRO {recovers  $\sigma$  at  $\tau=0$ , maximin as  $\tau \rightarrow \infty$ }
0:   Train  $\mu^{\text{new}} \leftarrow \text{BR}(\Pi, q(\tau))$ 
0:    $\Pi \leftarrow \Pi \cup \{\mu^{\text{new}}\}$ 
0:   Evaluate  $\mu^{\text{new}}$  against  $\Pi$  and update  $M$ 
0: end for
0: return  $\Pi, \sigma=0$ 
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and appended to the population:  $\Pi_i^{t+1} = \Pi_i^t \cup \{\mu_i^{\text{new}}\}$ . The full procedure is given in Algorithm 1.

The temperature  $\tau$  continuously interpolates between two known oracles. At  $\tau = 0$ ,  $q(\tau) = \sigma_{-i}^t$  and DF-PSRO recovers standard PSRO. As  $\tau \rightarrow \infty$ ,  $q(\tau)$  concentrates on  $\arg \min_j R_{bj}$ , recovering maximin over the current population. DF-PSRO therefore subsumes both oracles as limiting cases of a single robustness-controlled family.

### Theoretical Analysis

We establish two properties of DF-PSRO that together formalize its robustness–efficiency trade-off. Fix a player and let  $\Pi = \{\mu_1, \dots, \mu_n\}$  denote the current opponent population,  $\sigma \in \Delta(\Pi)$  the meta-Nash mixture, and  $\mu_b \in \Pi$  the strongest incumbent under  $\sigma$ . For a candidate policy  $\mu$  and opponent  $\mu_j$ , let  $R(\mu, j) = R(\mu, \mu_j)$ . Denote the standard PSRO oracle's solution by  $\mu^P = \arg \max_{\mu} \mathbb{E}_{\sigma}[R(\mu, \cdot)]$  and the DF-PSRO oracle's solution by  $\mu^{\text{DF}}(\tau) = \arg \max_{\mu} \mathbb{E}_{q(\tau)}[R(\mu, \cdot)]$ , where  $q(\tau)$  is given by Eq. (2). For  $\rho \geq 0$ , define the KL ambiguity set

$$\mathcal{Q}_{\rho} = \{q' \in \Delta(\Pi) : \text{KL}(q' \parallel \sigma) \leq \rho\}. \quad (4)$$

**Assumption 1** (Reference alignment). For any candidate  $\mu$ , the standardized payoff vectors of  $\mu$  and  $\mu_b$  are comonotone:  $\text{sign}(R(\mu, j) - R(\mu, k)) = \text{sign}(R(\mu_b, j) - R(\mu_b, k))$  for all  $j, k$ .

This assumption states that opponents that are difficult for the strongest incumbent  $\mu_b$  remain relatively difficult for any candidate  $\mu$ —an opponent-ranking consistency that holds when candidate policies share the same broad strategy class as  $\mu_b$ .

**Theorem 1** (Comparative robustness). Let  $\rho^*(\tau) = \text{KL}(q(\tau) \parallel \sigma)$ . Under Assumption 1, for any  $\rho \leq \rho^*(\tau)$ ,

$$\min_{q' \in \mathcal{Q}_{\rho}} \mathbb{E}_{q'}[R(\mu^{\text{DF}}(\tau), \cdot)] \geq \min_{q' \in \mathcal{Q}_{\rho}} \mathbb{E}_{q'}[R(\mu^P, \cdot)]. \quad (5)$$

*Proof sketch.* By construction,  $q(\tau)$  is the worst-case distribution within  $\mathcal{Q}_{\rho^*(\tau)}$  for the loss  $-R(\mu_b, \cdot)$ , which is the

closed-form solution of KL-constrained max. Under Assumption 1, this worst-case structure is preserved when the loss is replaced by  $-R(\mu, \cdot)$  for any candidate  $\mu$ , so  $q(\tau) \in \arg \min_{q' \in \mathcal{Q}_{\rho}} \mathbb{E}_{q'}[R(\mu, \cdot)]$  for any  $\rho \leq \rho^*(\tau)$ . Since  $\mu^{\text{DF}}(\tau)$  maximizes  $\mathbb{E}_{q(\tau)}[R(\mu, \cdot)]$  over candidates while  $\mu^P$  does not, the inequality follows from comparing the two BR objectives at the common worst-case distribution. The full proof is in Appendix .  $\square$

**Theorem 2** (Robustness–efficiency frontier). The DF-PSRO family  $\{\mu^{\text{DF}}(\tau) : \tau \geq 0\}$  traces a continuous curve in the  $(\text{Eff}, \text{Rob}_{\rho})$ -plane, where  $\text{Eff}(\mu) = \mathbb{E}_{\sigma}[R(\mu, \cdot)]$  and  $\text{Rob}_{\rho}(\mu) = \min_{q' \in \mathcal{Q}_{\rho}} \mathbb{E}_{q'}[R(\mu, \cdot)]$ . The endpoints satisfy:

- (i) when  $\tau = 0$ ,  $\mu^{\text{DF}}(0) = \mu^P$ , the standard PSRO best response;
- (ii) when  $\tau \rightarrow \infty$ ,  $\mu^{\text{DF}}(\tau) \rightarrow \arg \max_{\mu} R(\mu, j^*)$ , where  $j^* = \arg \min_j R(\mu_b, j)$ , recovering maximin-PSRO.

Moreover,  $\text{Eff}(\mu^{\text{DF}}(\tau))$  is non-increasing and  $\text{Rob}_{\rho}(\mu^{\text{DF}}(\tau))$  is non-decreasing in  $\tau$ .

*Proof sketch.* Continuity follows from continuity of  $q(\tau)$  in  $\tau$  (softmax structure) and continuity of the BR map under standard regularity. The endpoint at  $\tau = 0$  is immediate from  $q(0) = \sigma$ . The endpoint at  $\tau \rightarrow \infty$  follows from  $q(\tau) \rightarrow \delta_{j^*}$ , where  $j^*$  is determined by the reference  $\mu_b$ 's payoff vector. Monotonicity of  $\text{Eff}$  and  $\text{Rob}_{\rho}$  in  $\tau$  follows from the variational form of KL-DRO: increasing  $\tau$  shifts the BR objective toward worst-case opponents, which by definition cannot decrease robustness or increase nominal efficiency. Full proof in Appendix .  $\square$

Theorem 1 formalizes the claim that DF-PSRO is strictly more robust than standard PSRO under any KL-bounded opponent deviation. Theorem 2 positions DF-PSRO as a continuous interpolation between two known oracles: rather than choosing between standard PSRO's efficiency and maximin's robustness, practitioners select an operating point on the frontier via  $\tau$ .

### Residual Policy

In competitive multi-agent environments with fine-grained spatial observations, effective policy design requires agents to translate local spatial observations into concrete decisions that exploit positional advantages: identifying uncontested regions, adjusting behavior based on rival proximity, and allocating capacity toward high-value opportunities. This requires both stable spatial priors that capture general competitive structure and expressive learned components that refine decisions based on fine-grained local context.

Our framework addresses this through a residual policy design (Section ) that combines a feature-engineered macro controller with a learned U-Net micro policy, giving each agent the capacity to identify and exploit local spatial advantages. The DF-PSRO training scheme then provides the training stability required to realize these advantages consistently across diverse competitor configurations.

## Experiment

### Dataset and Simulator

We construct our simulation from three public sources: NYC TLC Trip Record Data (High-Volume FHV) for demand (NYC Taxi and Limousine Commission 2026b), OpenStreetMap for the Manhattan road network (OpenStreetMap contributors 2026), and NYC TLC Taxi Zones for spatial anchoring (NYC Taxi and Limousine Commission 2026a). Trip records are aggregated into 5-minute decision intervals, and zone-level demand is mapped to network nodes via the road graph clipped to Manhattan. To obtain a smooth demand process, we fit per-day dual GMMs (separate pickup and dropoff) in  $(x, y, t)$  space, with each day’s model trained on a wrapped temporal window covering the current day plus the adjacent half-days to reduce midnight boundary artifacts. At each simulation epoch, demand is sampled from the corresponding day-specific mixture.

We evaluate all methods in a competitive ride-hailing simulator over a  $20 \times 20$  grid road network. At each epoch,  $K$  companies each operate a fleet of  $N = 40$  autonomous vehicles. Traveler demand is generated by a synthetic Gaussian mixture process with fixed pickup and drop-off distributions. The matching mechanism follows the transition dynamics in Section . Unless otherwise stated, experiments use  $K = 2$  competing companies, a sampling intensity of 80 travelers per epoch, and a simulation horizon of  $T = 200$  evaluation epochs preceded by 500 warm-up epochs.

### Robustness to Opponent Strategies

To assess robustness beyond the within-distribution training regime, we evaluate three oracle variants (Standard PSRO, Maximin-PSRO, DF-PSRO) under one normal and three adversarial settings. The **normal setting** is symmetric: both companies operate fleets of equal size, face customers without brand preference, and price through their trained policies without external perturbation. The three adversarial settings each perturb a single dimension of this baseline: (i) **Aggressive undercutting**: the opponent quotes prices 50% below the level produced by its trained policy; (ii) **Fleet asymmetry**: the opponent operates a fleet 50% larger than ours; (iii) **Brand-biased demand**: customers exhibit a fixed preference for the opponent at equal posted prices. For each setting, we report each method’s revenue as a percentage change relative to its own revenue in the normal setting, isolating the impact of the perturbation from absolute-revenue differences across methods.

### Feature Importance

Starting from the full FeatureEng model trained under DF-PSRO, we ablate each spatial feature by setting its contribution to zero: (a) **no  $F_{\text{rival}}$** : competitor positions removed, (b) **no  $F_{\text{trav}}$** : traveler density removed, (c) **no  $F_{\text{center}}$** : city-center bias removed. We also isolate the effect per decision component (pricing, willingness, relocation).

Figure XX reports revenue drop relative to the full model. Removing  $F_{\text{rival}}$  causes the largest decline across all three decision components. Its effect is strongest in pricing and

relocation: without competitor information, the vehicle cannot detect uncontested zones to raise prices, nor avoid competitor-saturated regions when relocating. Removing  $F_{\text{trav}}$  reduces revenue primarily through the pricing component. Removing  $F_{\text{center}}$  has the smallest standalone effect but causes vehicles to over-relocate to peripheral, low-demand areas.

## Conclusion

Competitive markets rarely play Nash. The premise of this paper is that population-based learning should reflect that fact: if off-equilibrium behavior is a feature of the environment rather than a corner case, then robustness to such behavior belongs inside the oracle, not bolted on afterward. DF-PSRO is one way to do this: a soft KL reweighting anchored on the strongest incumbent, controlled by a single temperature that exposes the robustness–efficiency trade-off as a tunable axis rather than a binary choice between PSRO and maximin. The street-level ride-hailing results are consistent with this view: robustness gains appear precisely where standard PSRO’s equilibrium assumption breaks, and disappear where it holds.

Several caveats temper these claims. The comparative guarantee leans on a co-monotonicity assumption between candidate and incumbent payoffs, which is natural when policies share a strategy class but harder to justify in genuinely heterogeneous populations. Anchoring on the strongest incumbent keeps per-iteration cost flat but is a surrogate, not the true candidate-aware worst case. And  $\tau$  is still a knob—scale-invariant, but not self-tuning. We see the most interesting next steps as making  $\tau$  respond to observed opponent behavior, swapping KL for divergences that capture different notions of deviation, and testing whether the same picture holds beyond two operators and beyond ride-hailing.

## Appendix A

We model the competitive ride-hailing system as a partially observable Markov game (POMG)  $\mathcal{M} = \langle \mathcal{K}, \mathcal{S}, \{\mathcal{A}_k\}_{k \in \mathcal{K}}, \mathcal{P}, \{R_k\}_{k \in \mathcal{K}}, \{\Omega_k\}_{k \in \mathcal{K}}, \{\mathcal{O}_k\}_{k \in \mathcal{K}}, T \rangle$ , where  $\mathcal{K} = \{1, \dots, K\}$  is the set of competing companies,  $\mathcal{S}$  is the global state space,  $\mathcal{A}_k$  is the action space of company  $k$ ,  $\mathcal{P} : \mathcal{S} \times \prod_k \mathcal{A}_k \rightarrow \Delta(\mathcal{S})$  is the transition kernel,  $R_k$  is the reward function of company  $k$ ,  $\Omega_k$  is its observation space with observation function  $\mathcal{O}_k : \mathcal{S} \rightarrow \Omega_k$ , and  $T$  is the finite time horizon. Each company  $k$  operates a fleet of  $N_k$  AVs over a shared road network  $G = (\mathcal{V}, \mathcal{E})$ .

### State and Observation

At each epoch  $t$ , the global state  $s_t \in \mathcal{S}$  consists of the positions and operational statuses of all AVs across all companies, together with the set of active travelers  $\mathcal{J}_t$ . Each company observes only a partial view of the global state:  $o_{t,k} = \mathcal{O}_k(s_t) \in \Omega_k$ .

In our per-vehicle formulation, each vehicle receives a structured local observation

$$o_{t,k,v} = (s_v, \mathcal{V}^{\text{same}}, \mathcal{V}^{\text{rival}}, \mathcal{H}^{\text{rival}}, \mathcal{J}^{\text{pick}}, \mathcal{J}^{\text{unpick}}, \tau)$$

where  $s_v$  denotes self state.  $\mathcal{V}^{\text{same}}$  and  $\mathcal{V}^{\text{rival}}$  represent the sets of same-company and other-company (rival) AVs, respectively, and  $\mathcal{H}^{\text{rival}}$  denotes rival human-driven vehicles (HDVs).  $\mathcal{J}^{\text{pick}}$  and  $\mathcal{J}^{\text{unpick}}$  denote the sets of pickable and unpickable travelers. The scalar  $\tau \in [0, 1]$  represents normalized time-of-day. All entities are encoded using normalized spatial coordinates.

We consider two information regimes: (i) a *zone-level* observation with coarse aggregated features, and (ii) a *street-level* observation with exact fine-grained spatial positions. This distinction determines whether policies can exploit local spatial asymmetries.

### Action

Each vehicle  $v$  belonging to company  $k$  at every epoch

$$a_{t,k,v} = (g_{t,v}, p_{t,v}, \mathbf{w}_{t,v}),$$

where  $g_{t,v} \in \mathcal{G}$  is a relocation destination on a discrete grid,  $p_{t,v} \in \mathbb{R}_{>0}$  is the quoted unit-distance price, and  $\mathbf{w}_{t,v} \in \mathbb{R}^{|\mathcal{J}^{\text{pick}}|}$  is the willingness vector aligned with the company-specific pickable traveler list  $\mathcal{J}_{t,k}^{\text{pick}}$ . The company-level action is  $a_{t,k} = \{a_{t,k,v}\}_{v=1}^{N_k}$ . The policy always predicts all three components each epoch; the simulator executes the components that are relevant to the vehicle's current operational status.

### Transition

The system evolves according to the joint action profile  $(a_{t,1}, \dots, a_{t,K})$  through three coupled mechanisms:

- (i) **Traveler-company selection.** Each traveler  $j \in \mathcal{J}_t$  first decides whether to take a ride, then selects a company. For company  $k$ , traveler  $j$  observes an aggregated unit-distance price

$$\bar{p}_{t,j,k} = \frac{\sum_{v \in \mathcal{F}_{t,k}} w_{t,j,v} p_{t,v}^t}{\sum_{v \in \mathcal{F}_{t,k}} w_{t,j,v}}, \quad w_{t,j,v} = \frac{1}{(d_{t,j,v})^2 + \varepsilon},$$

where  $d_{t,j,v}$  is pickup distance from vehicle  $v$  to traveler  $j$ ,  $\mathcal{F}_{t,k}$  is available vehicles in company  $k$ , and  $\varepsilon = 10^{-12}$ . Let  $k^* = \arg \min_k \bar{p}_{t,j,k}$ , and the ride utility is

$$U_j^{\text{ride}} = \alpha - \beta \left( \frac{\bar{p}_{t,j,k^*}}{p_{\text{ref}}} - 1 \right),$$

with traveler participation probability  $\sigma(U_j^{\text{ride}})$ . Conditional on participation, company choice is

$$P(k | j) = \frac{\exp(-\gamma(\bar{p}_{t,j,k}/p_{\text{ref}} - 1))}{\sum_{k'} \exp(-\gamma(\bar{p}_{t,j,k'}/p_{\text{ref}} - 1))}.$$

- (ii) **Company-traveler matching.** Within selected company  $k$ , travelers are assigned to vehicles via deferred acceptance driven by willingness scores. Each traveler proposes to

$$v^* = \arg \max_{v \in \mathcal{F}_{t,k}} w_{t,j,v},$$

vehicles tentatively keep their highest-score proposal and reject others, and rejected travelers re-propose until convergence, yielding stable matching  $\mathcal{M}_{t,k}$ , which is the set of vehicle-traveler pairs matched for company  $k$  at epoch  $t$ .

- (iii) **Vehicle movement.** Matched vehicles execute pickups and complete trips on the network  $G$ . Unmatched vehicles relocate toward their chosen target  $g_v$ .

Equivalently, the transition can be written as

$$s_{t+1} \sim \mathcal{P}(\cdot | s_t, \mathbf{a}_t),$$

where stochasticity comes from traveler participation, company selection, relocation sampling within a grid cell, and traffic/trip-time realization.

### Reward

The reward for company  $k$  at epoch  $t$  is the total trip revenue collected from completed matches:

$$R_k(s_t, \mathbf{a}_t) = \sum_{(v,j) \in \mathcal{M}_{t,k}} p_v \cdot \text{dist}(o_j, d_j),$$

where  $\text{dist}(o_j, d_j)$  is the network distance from traveler  $j$ 's origin  $o_j$  to destination  $d_j$ . Each company seeks to maximize its cumulative reward  $\sum_{t=1}^T R_k(s_t, \mathbf{a}_t)$ .

## Appendix B

### Proof of Theorem 1

$q(\tau)$  as the worst-case distribution under reference loss. Define the standardized reference loss  $\ell_j^b = -(R(\mu_b, j) - \bar{R}_b)/s_{R_b}$ . Consider the KL-constrained maximization problem

$$\max_{q' \in \Delta(\Pi)} \mathbb{E}_{q'}[\ell^b] \quad \text{s.t.} \quad \text{KL}(q' \| \sigma) \leq \rho. \quad (6)$$

The Lagrangian is

$$\mathcal{L}(q', \lambda, \nu) = \sum_j q'_j \ell_j^b - \lambda \left( \sum_j q'_j \log \frac{q'_j}{\sigma_j} - \rho \right) - \nu \left( \sum_j q'_j - 1 \right), \quad (7)$$

with  $\lambda \geq 0$ . Setting  $\partial \mathcal{L} / \partial q'_j = 0$  yields

$$q_j^{*'}(\lambda) = \frac{\sigma_j \exp(\ell_j^b / \lambda)}{\sum_k \sigma_k \exp(\ell_k^b / \lambda)}. \quad (8)$$

Setting  $\lambda = 1/\tau$  recovers  $q(\tau)$  in Eq. (2) exactly. The KL radius achieved by  $q(\tau)$  is

$$\rho^*(\tau) := \text{KL}(q(\tau) \| \sigma), \quad (9)$$

so  $q(\tau)$  is the unique optimal solution of (6) at  $\rho = \rho^*(\tau)$ .

**Worst-case structure transfers to candidate losses (Assumption 1).** For any candidate policy  $\mu$ , define the standardized loss  $\ell_j^\mu = -(R(\mu, j) - \bar{R}_\mu)/s_{R_\mu}$ . By Assumption 1,  $\ell^\mu$  and  $\ell^b$  are co-monotone: their components agree in ordering across  $j$ .

**Lemma 1.** If  $\ell^\mu$  and  $\ell^b$  are co-monotone, then

$$\arg \max_{q' \in \mathcal{Q}_\rho} \mathbb{E}_{q'}[\ell^\mu] = \arg \max_{q' \in \mathcal{Q}_\rho} \mathbb{E}_{q'}[\ell^b] = q(\tau) \quad (10)$$

for  $\rho = \rho^*(\tau)$ .

*Proof of Lemma 1.* The KL-constrained maximizer concentrates probability mass on indices with the highest loss values, with the exponential-tilt profile fixed by  $\rho$  (Step 1). Since  $\ell^\mu$  and  $\ell^b$  have the same ranking by co-monotonicity, the relative mass profile that maximizes  $\mathbb{E}_{q'}[\ell^\mu]$  matches that which maximizes  $\mathbb{E}_{q'}[\ell^b]$ . The closed-form solution is the same.  $\square$

Hence, for any  $\rho \leq \rho^*(\tau)$ , the worst-case distribution under candidate loss  $\ell^\mu$  within  $\mathcal{Q}_\rho$  is upper-bounded in expectation by  $\mathbb{E}_{q(\tau)}[\ell^\mu]$ , and equality holds at  $\rho = \rho^*(\tau)$ .

### Comparing DF-PSRO and PSRO worst-case payoffs.

Translating from loss to payoff:  $\min_{q' \in \mathcal{Q}_\rho} \mathbb{E}_{q'}[R(\mu, \cdot)] = \mathbb{E}_{q(\tau)}[R(\mu, \cdot)]$  at  $\rho = \rho^*(\tau)$ , and is lower-bounded by  $\mathbb{E}_{q(\tau)}[R(\mu, \cdot)]$  for any  $\rho \leq \rho^*(\tau)$  (since smaller  $\rho$  admits fewer distributions, raising the min).

The DF-PSRO oracle satisfies, by definition,

$$\mathbb{E}_{q(\tau)}[R(\mu^{\text{DF}}(\tau), \cdot)] \geq \mathbb{E}_{q(\tau)}[R(\mu^{\text{P}}, \cdot)]. \quad (11)$$

Combining with the worst-case identity above,

$$\min_{q' \in \mathcal{Q}_\rho} \mathbb{E}_{q'}[R(\mu^{\text{DF}}(\tau), \cdot)] \geq \mathbb{E}_{q(\tau)}[R(\mu^{\text{DF}}(\tau), \cdot)] \quad (12)$$

$$\geq \mathbb{E}_{q(\tau)}[R(\mu^{\text{P}}, \cdot)] \quad (13)$$

$$\geq \min_{q' \in \mathcal{Q}_\rho} \mathbb{E}_{q'}[R(\mu^{\text{P}}, \cdot)]. \quad (14)$$

The last inequality uses the same worst-case structure applied to  $\mu^{\text{P}}$ .  $\square$

### Proof of Theorem 2

We prove continuity, the two endpoints, and the monotonicity properties separately.

**Continuity.** The reweighting map  $\tau \mapsto q(\tau)$  is continuous on  $\tau \in [0, \infty)$  as a composition of the exponential tilt  $\tau \mapsto \sigma_j \exp(\tau \ell_j^b)$  and softmax normalization, both continuous. Under standard regularity of the BR oracle (compact candidate set, continuous payoff function), the BR map  $q \mapsto \arg \max_\mu \mathbb{E}_q[R(\mu, \cdot)]$  is continuous, so  $\tau \mapsto \mu^{\text{DF}}(\tau)$  is continuous. Both Eff and  $\text{Rob}_\rho$  are continuous functionals of  $\mu$ , yielding continuity of the frontier curve.

**Endpoint at  $\tau = 0$ .** At  $\tau = 0$ ,  $\exp(0) = 1$  for all  $j$ , so  $q(0) = \sigma$ . The DF-PSRO oracle reduces to  $\arg \max_\mu \mathbb{E}_\sigma[R(\mu, \cdot)] = \mu^{\text{P}}$ .

**Endpoint at  $\tau \rightarrow \infty$ .** As  $\tau \rightarrow \infty$ , the exponential tilt  $\exp(\tau \ell_j^b) = \exp(-\tau \cdot (R(\mu_b, j) - \bar{R}_b)/s_{R_b})$  is dominated by  $j^* = \arg \min_j R(\mu_b, j)$ . By standard softmax limit,  $q(\tau) \rightarrow \delta_{j^*}$ . Therefore

$$\mu^{\text{DF}}(\tau) \rightarrow \arg \max_\mu \mathbb{E}_{\delta_{j^*}}[R(\mu, \cdot)] = \arg \max_\mu R(\mu, j^*), \quad (15)$$

which is the maximin best response against the single hardest opponent for  $\mu_b$  in  $\Pi$ .

**Monotonicity of Eff and  $\text{Rob}_\rho$ .** Eff is non-increasing in  $\tau$ : At  $\tau = 0$ ,  $\mu^{\text{DF}}(0) = \mu^{\text{P}}$  which by definition maximizes Eff. For any  $\tau > 0$ ,  $\text{Eff}(\mu^{\text{DF}}(\tau)) \leq \text{Eff}(\mu^{\text{P}})$ .

$\text{Rob}_\rho$  is non-decreasing in  $\tau$  (for fixed  $\rho$ ): By Theorem 1 applied with reference  $\mu^{\text{DF}}(\tau)$  versus  $\mu^{\text{DF}}(\tau')$  for  $\tau' < \tau$ , the policy at higher  $\tau$  achieves a tighter (larger) lower bound on worst-case payoff over  $\mathcal{Q}_\rho$ .  $\square$

### Residual Policy Details

**Per-Vehicle Formulation** Our policy is defined at the per-vehicle level. At each epoch, vehicle  $v$  receives a local structured observation  $o_{t,k,v}$  and outputs  $(g_{t,v}, p_{t,v}, \mathbf{w}_{t,v})$ . All vehicles share one policy parameterization, so execution is per-vehicle while learning remains statistically efficient.

**Feature-Engineered Base Policy** The base policy uses additive biases and distance-based falloff terms (exponential or squared-exponential) to convert local geometry into decisions.

#### Pricing head.

- *Rival pressure*: nearby rivals reduce local market power, modeled by distance-weighted rival density around the vehicle.
- *Traveler density*: nearby traveler concentration raises local demand, modeled by distance-weighted traveler density.
- *Center attractiveness*: proximity to city center captures systematic demand gradients.

These factors are combined linearly with a bias to produce unit-distance quote  $p_v$ .

#### Willingness head.

- *Pickup proximity*: closer traveler origins increase willingness.
- *Traveler posted/committed price*: higher observed price raises willingness through a monotone transform.
- *Local congestion at traveler origin*: dense nearby vehicles reduce willingness.
- *Destination attractiveness*: destinations closer to the city core increase willingness.

These terms are combined with a bias to obtain traveler-specific score  $w_{v,j}$ .

#### Relocation head.

- *Traveler-driven proposal map*: a Gaussian kernel density over traveler locations forms the candidate relocation demand map.
- *Vehicle-avoidance term*: cells near other vehicles are down-weighted by distance-based repulsion.
- *Center regularization*: an additional center-distance term controls central over-concentration.

The final relocation distribution multiplies these components and normalizes over valid cells.

**U-Net Residual Branch** The U-net branch (Figure 3) consumes rasterized street-level observations and predicts residual corrections to price, willingness, and relocation logits. We choose U-net because skip connections preserve local geometric detail while the encoder captures broader competitive context.



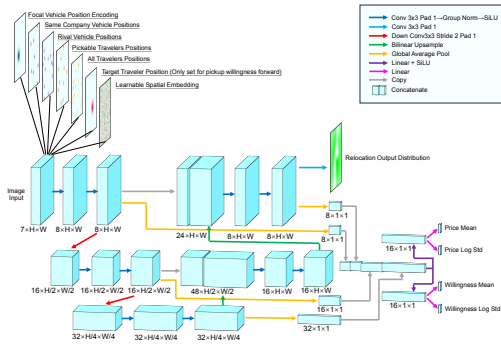


Figure 3: U-Net architecture for microscopic spatial pattern learning and action fine-tuning. The model takes as input a multi-channel image encoding the vehicle’s observation. It produces a relocation log-probability distribution from the decoder output, along with price and traveler pickup–willingness distributions parameterized as Gaussian means and log standard deviations via two separate output heads.

**REINFORCE Training** Given rollout return  $r$ , we use normalized advantage  $\hat{A} = \frac{r-b}{|b|+\epsilon}$ , with entropy regularization to prevent premature collapse. The baseline  $b$  is the round-average per-step reward across workers in each best-response round. This baseline is a good estimator because it averages over diverse parallel rollouts against the current opponent mixture, reducing variance while staying aligned with the current training distribution.

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